Fiscal Year:	FY 2016	**Task Last Updated:**	FY 01/12/2016
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Project Title:	Targeting Protein Cross Talk Signaling as a Biological Predictor for Space Radiation Cancer Risk		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Zip Code:	78229-3901	**Congressional District:**	21
Comments:			
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Flight Program:			
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Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Blakely, Eleanor (Lawrence Berkeley National Laboratory) Mohan, Sumathy (University of Texas, San Antonio) Prihoda, Tom (University of Texas, San Antonio)		
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Task Description:	One of the important concerns of 'late effects' after heavy ion radiation exposure is the onset of solid cancers. Earlier mechanistic studies independently reported by several investigators emphasized intercellular signaling as a key aspect of carcinogeneis through oxidative damage, genomic instability, and inflammation. Alterations in the architectural network of the cellular signaling mechanism by modifying the functional aspects of signaling mediators and fixation of those alterations may translate into oncogenic processes leading to carcinogenesis. However, it is not yet known how these mediators are orchestrated to result in the induction of cancer. In cases such as low fluxes of heavy ion radiation, where only less irradiated cells (primarily at low fluences) co-exist with more nonirradiated counter parts, the alterations may be predominantly through an intercellular communication between irradiated to non-targeted cells than merely through the irradiated cells themselves. Therefore non-targeted effect (bystander effect) is considered to be one of the main factors for cancer initiation. It is highly unlikely that an independent factor or involvement of a unidirectional signaling pathway (linear or non-cyclic pathway), as reported in most of the radiation-induced bystander studies, can provide answers to alleviate cancer risk uncertainties. Unless the signals are recycled and amplified, it is possible that an intact homeostatic mechanism may recognize and dissipate these signals, which flow from point 'A' to point 'B' and weaken at every downstream step. Further, altered cells may be prompted to undergo apoptosis and be removed from the system before malignant transformation. Therefore, these cells are unlikely to cause any late effects. We therefore hypothesized that radiation-triggered signals are sustained through a positive-feedback loop, amplified and transmitted to the surrounding tissue through a bystander mechanism, resulting in replication and radial expansion of those signals. This sequence of events initiates cancer pre-disposing factors, genomic instability, and survival advantage. These factors are responsible for clonal expansion of cells that are prone to be tumorigenic. Two key proteins, nuclear factor kappa B (NF-kB) and tumor necrosis factor alpha (TNF-a), sequentially co-ordinate and control this cascade of events. The following specific aims are proposed to test the hypothesis proposed. Aim 1: To study the significance of low doses (0, 0.2, 0.4, or 0.8 Gy) of high-LET (linear energy transfer) radiation exposure on an altered autocrine signaling mechanism that initiates, amplifies, and sustains a positive feedback cycle, i.e., triggering NF-kB -->TNF-a --> NF-kB signaling in the targeted cells. Aim 2: To study and compare the significance of the inter-cellular communication (through paracrine mechanism) that is responsible for amplification of the signals by replicating the activation of NF-kB - TNF-a positive feedback loop in the non-targeted bystander cells. Aim 3: To study, in targeted versus non-targeted bystander cells, the influence of sustained amplification of NF-kB - TNF-a positive feedback loop on (a) stimulating predisposing factors for carcinogenesis and (b) potentiating in vitro and in vivo tumorigenesis. In aim 1, we compared the outcomes with regard to the LETs (51, and 181 keV/um) of two different HZE (high energy particle) ion beams (28Si and 56Fe) accelerated to the same velocity (600 MeV/amu), and having similar track structure dimensions, but different ionization densities. This was examined in normal human endothelial cells maintained in their micro-environmental niche. In aim 2, the sustained amplification of those initial radiation-induced signals were examined. In aim 3, we investigated, whether or not, the predisposing factors for carcinogenesis proposed in this study are the concurrent development of genomic instability (mediated by TNF-a induced free radical mechanism) and acquired survival advantage (enhanced by sustained NF-kB activation). The carcinogenesis end point were assessed by neoplastic transformation potentials, and the ability to grow palpable tumor(s) in nude mice.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study will clarify whether radiation-induced molecular changes that are persistently maintained through positive feed-back cycle through two key proteins, nuclear factor kappa B (NF-kB) and tumor necrosis factor alpha (TNF-a), are amplified over time. These altered molecular events are responsible for clonal expansion of cells that are prone to be tumorigenic. In terms of usefulness of the above study on Earth, normal tissue response is one of the major focuses to achieve acceptable therapeutic gain in radiotherapy of many cancers. Identification and characterization of the molecular mediators that play a role in radiation-induced bystander effects that could lead to genomic instability and survival advantage after therapeutic doses of low LET radiation is important to understand the radiation-induced delayed effect in normal tissue. Therefore the findings from the study proposed with space radiation can set a stage to understand the underlying mechanism of tumorigenesis, tumor recurrence, and tumor growth at the treatment site.
	Space travelers are irradiated with cosmic rays to a dose rate considerably higher than that received on Earth. At the dose and dose rates encountered in space, the important risk is the induction of cancer in the astronauts. Earlier studies mostly from experiments observing chromosome aberrations and gene mutations clearly suggested the need for understanding the cellular mechanisms in order to clarify the uncertainties in cancer risk. The hypothesis we tested in this project was that (i) whether radiation-triggered signals are sustained through a positive-feedback loop, amplified, and transmitted to the surrounding tissue through a bystander mechanism, resulting in replication and radial expansion of those signals; and (ii) whether this sequence of events initiates cancer pre-disposing factors, genomic instability, and survival advantage. In addition we investigated whether these factors are responsible for clonal expansion of cells that are prone to be tumorigenic. The specific aims originally proposed in this three year study are: Aim 1: To study the significance of low doses (0, 0.2, 0.4, or 0.8 Gy) of high-LET radiation exposure on an altered autocrine signaling mechanism that initiates, amplifies, and sustains a positive feedback cycle, i.e., triggering NF-kB -->TNF-a --> NF-kB signaling in the targeted cells. Aim 2: To study and compare the significance of the inter-cellular communication (through paracrine mechanism) that is responsible for amplification of the signals by replicating the activation of NF-kB - TNF-a positive feedback loop in the non-targeted bystander cells. Aim 3: To study, in targeted versus non-targeted bystander cells, the influence of sustained amplification of NF-kB - TNF-a positive feedback loop on (a) stimulating predisposing factors for carcinogenesis and (b) potentiating in vitro and in vivo tumorigenesis. In aim 1, we proposed to compare the outcomes with regard to the LETs (51 and 181 keV/um) of two different HZE ion beams (28Si and 56Fe) accelerated to the same velocity (600 MeV/ amu), and having similar track structure dimensions, but different ionization densities. This was examined in normal human endothelial cells maintained in their micro-environmental niche. In aim 2, we proposed to determine whether the sustained amplification of those initial radiation-induced signals were examined. In aim 3, we investigated the predisposing factors for carcinogenesis proposed in this study are the concurrent development of

Task Progress:	genomic instability (mediated by TNF-a induced free radical mechanism) and acquired survival advantage (enhanced by sustained NF-kB activation). The carcinogenesis end points were assessed by neoplastic transformation potentials, and the ability to grow palpable tumor(s) in nude mice. None of the specific aims were changed In this progress report, we summarize the research work conducted during the no cost extension period covering the last six months. During this period, a substantial time was devoted to consolidate/revise and resubmit all the data obtained through the collaborative initiatives. Four peer reviewed journal articles were published after the last report [Breast Cancer Res Treat. 150(3):487-99; BMC-Cancer. Jul 10;15:514. doi: 10.1186/s12885-015-1463-y; BMC Genomics. Jul 7;16:501. doi: 10.1186/s12864-015-1642; and PLoS ONE. 10(7):e0133016. PMID: 26208353]. Also, time was devoted to compile the data for a new manuscript on 'Novel mechanism of sustained space radiation bystander effect.' The manuscript is in preparation and will be submitted in spring 2016. In addition, during this period, experiments were conducted with pro-tumorigenic cells (BEP2D, and MCF-10A) that are either irradiated with 56Fe or 28Si (600 MeV/amu) to a dose of 0.2 Gy x 3 or mock irradiated. The cells were repeatedly cultured to 30 to 40 passages after radiation exposure and injected into nude mice to determine the tumorigenic potentials. Similarly, the cells pre-incubated with NF-kB inhibitor, isohelanin, for 30 minutes prior to exposure were also, in parallel, cultured to subsequent passages. Cells at a density of 10 x 10(6) cells in 0.2 ml saline were injected into nude mice. In a second set of experiments, the pro-tumorigenic cells were irradiated and co-cultured with non-irradiated pro-tumorigenic cells. Non-targeted pro-tumorigenic cells that are co-cultured with the irradiated tumor cells were continuously passaged to 30-40 passages. The later experiments were conducted in order to identify whether the TNF-a secreted out from the irradiated cells are responsible for initiating a similar TNF-a-NF-kB feedback cycle in the bystander cells and those cells are capable of transforming into tumorigenic cells. A total of 5 nude mice (Foxn1nu/Foxn1nu) at a 6-8 weeks of age per group was used in this study. The first set of results indicated that the pro-tumorigenic cells co-cultured with irradiated cells were capable of transforming into tumorigenic cells. Both group of mice, i.e., mice either receiving cells that were directly irradiated or mice that were injected with bystander cells that were co-cultured with irradiated cells, showed palpable tumors on day 5. Also, during this period, TNFRI and TNFRII knockout mice were bred successfully. A total of 12 mice at 6-8 weeks old were used to isolate aortic endothelial cells from thoracic aortae. These cells were used to understand the role of TNF-a in the space radiation- and low LET radiation-induced bystander mechanism. As mentioned above the data has been published in four research articles and the current data will be compiled and submitted as a full length manuscript in spring 2016.
Bibliography Type:	Description: (Last Updated: 04/11/2021)
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